CardioSens: An Advanced Technology for Heart Abnormalities Detection and Diagnosis

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Under the guidance of

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Submitted in partial fulfillment of the requirement of the Degree of Bachelor of Technology
of Maulana Abul Kalam Azad University (formerly West Bengal University of Technology), Kolkata

Department of Electronics and Communication Engineering

St. Thomas' College of Engineering and Technology

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Vision of the Department

- To build a strong teaching and research environment to cater to the manpower needs in Industrial and Academic domains of the rapidly growing Electronics and Communication Engineering.

Mission of the Department

- To produce certified industry-ready professionals in Electronics and Communication Engineering, though innovative educational programs incorporating laboratory practices and project-based teaching-learning processes, in a modern environment.
- To create knowledge base of advanced technologies through research in the area of Electronics and Communication, for competitive and sustainable development of the country.
- To groom the department as a learning center to inculcate advancement of technology in Electronics and Communication Engineering with social values and environmental awareness.

Program Specific Outcome (PSOs)

After completion of program graduate engineer would have:

- **PSO1. Professional skills:** An ability to apply the knowledge in Electronics and Communication Engineering in various areas, like Communications, Signal processing, VLSI and Embedded Systems.
- **PSO2. Competency:** An ability to qualify at the State, National and International level competitive examinations for employment, higher studies and research

Program Outcome

Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society**: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability**: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and teamwork**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning**: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Course Outcomes for Final Year Project

After completion of the project, student will be able to:

CO no.	Outcome Statements	Bloom's Level
CO1	Apply technical knowledge in the solution of complex real-life problems related to public health and safety, culture, society, and environment	3
CO2	Review research literature and use the research-based knowledge to identify, formulate, and analyze the problem	5
CO3	Design innovative solutions for complex engineering problems , which will be published as research paper or developed as a marketable product in future	6
CO4	Apply modern tools to design and develop the solution to problem	3
CO5	Assess societal, health, safety, legal and cultural issues relevant to project and the responsibilities relevant to the professional engineering practice.	5
CO6	Evaluate the impact on the environment of working project	5
CO7	Apply ethical principles and commit to ethics and responsibilities related to engineering practice.	3
CO8	Function effectively as a member or leader of a team	3
CO9	Communicate effectively on professional activities with the team members, superiors and with society at large	3
C10	Plan , manage the project and control finance as a member and leader in a team.	3
C11	Apply the knowledge acquired during the project, in future higher studies or professional job.	4

Bloom's Level: Remember = 1, Understand = 2, Apply =3 Analyze = 4 Evaluate = 5, Create = 6

Program Outcomes -Competencies – Performance Indicators (PIs)

PO 1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems.	
Competency	**Indicators**
1.1 Demonstrate competence in mathematical modelling	1.1.1 Apply mathematical techniques such as calculus, linear algebra, and statistics to solve problems 1.1.2 Apply advanced mathematical techniques to model and solve Electronics engineering problems
1.2 Demonstrate competence in basic sciences	1.2.1 Apply laws of natural science to an engineering problem
1.3 Demonstrate competence in engineering fundamentals	1.3.1 Apply fundamental engineering concepts to solve engineering problems
1.4 Demonstrate competence in specialized engineering knowledge to the program	1.4.1 Apply Electronics engineering concepts to solve engineering problems.

PO 2: Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.	
Competency	**Indicators**
2.1 Demonstrate an ability to identify and formulate complex engineering problem	2.1.1 Articulate problem statements and identify objectives 2.1.2 Identify engineering systems, variables, and parameters to solve the problems 2.1.3 Identify the mathematical, engineering and other relevant knowledge that applies to a given problem
2.2 Demonstrate an ability to formulate a solution plan and methodology for an engineering problem	2.2.1 Reframe complex problems into interconnected sub-problems 2.2.2 Identify, assemble and evaluate information and resources. 2.2.3 Identify existing processes/solution methods for solving the problem, including forming justified approximations and assumptions 2.2.4 Compare and contrast alternative solution processes to select the best process.
2.3 Demonstrate an ability to formulate and interpret a model	2.3.1 Combine scientific principles and engineering concepts to formulate model/s (mathematical or otherwise) of a system or process that is appropriate in terms of applicability and required accuracy. 2.3.2 Identify assumptions (mathematical and physical) necessary to allow modelling of a system at the level of accuracy required
2.4 Demonstrate an ability to execute a solution process and analyze results	2.4.1 Apply engineering mathematics and computations to solve mathematical models 2.4.2 Produce and validate results through skillful use of contemporary engineering tools and models 2.4.3 Identify sources of error in the solution process, and limitations of the solution. 2.4.4 Extract desired understanding and conclusions consistent with objectives and limitations of the analysis

PO 3: Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations. .	
Competency	**Indicators**
3.1 Demonstrate an ability to define a complex/ open-ended problem in engineering terms	3.1.1 Recognize that need analysis is key to good problem definition 3.1.2 Elicit and document, engineering requirements from stakeholders 3.1.3 Synthesize engineering requirements from a review of the state-of-the-art 3.1.4 Extract engineering requirements from relevant engineering Codes and Standards such as ASME, ASTM, BIS, ISO and ASHRAE. 3.1.5 Explore and synthesize engineering requirements considering health, safety risks, environmental, cultural and societal issues 3.1.6 Determine design objectives, functional requirements and arrive at specifications
3.2 Demonstrate an ability to generate a diverse set of alternative design solutions	3.2.1 Apply formal idea generation tools to develop multiple engineering design solutions 3.2.2 Build models/prototypes to develop a diverse set of design solutions 3.2.3 Identify suitable criteria for the evaluation of alternate design solutions.
3.3 Demonstrate an ability to select an optimal design scheme for further development	3.3.1 Apply formal decision-making tools to select optimal engineering design solutions for further development 3.3.2 Consult with domain experts and stakeholders to select candidate engineering design solution for further development
3.4 Demonstrate an ability to advance an engineering design to defined end state	3.4.1 Refine a conceptual design into a detailed design within the existing constraints (of the resources) 3.4.2 Generate information through appropriate tests to improve or revise the design

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.	
Competency	**Indicators**
4.1 Demonstrate an ability to conduct investigations of technical issues consistent with their level of knowledge and understanding	4.1.1 Define a problem, its scope and importance for purposes of investigation 4.1.2 Examine the relevant methods, tools and techniques of experiment design, system calibration, data acquisition, analysis and presentation 4.1.3 Apply appropriate instrumentation and/or software tools to make measurements of physical quantities 4.1.4 Establish a relationship between measured data and underlying physical principles
4.2 Demonstrate an ability to design experiments to solve open-ended problems	4.2.1 Design and develop an experimental approach, specify appropriate equipment and procedures 4.2.2 Understand the importance of the statistical design of experiments and choose an appropriate experimental design plan based on the study objectives
4.3 Demonstrate an ability to analyze data and reach a valid conclusion	4.3.1 Use appropriate procedures, tools and techniques to conduct experiments and collect data 4.3.2 Analyze data for trends and correlations, stating possible errors and limitations 4.3.3 Represent data (in tabular and/or graphical forms) so as to facilitate analysis and explanation of the data, and drawing of conclusions 4.3.4 Synthesize information and knowledge about the problem from the raw data to reach appropriate conclusions

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.	
Competency	**Indicators**
5.1 Demonstrate an ability to identify / create modern engineering tools, techniques and resources	5.1.1 Identify modern engineering tools such as computer-aided drafting, modeling and analysis; techniques and resources for engineering activities 5.1.2 Create/adapt/modify/extend tools and techniques to solve engineering problems
5.2 Demonstrate an ability to select and apply discipline-specific tools, techniques and resources	5.2.1 Identify the strengths and limitations of tools for (i) acquiring information, (ii) modeling and simulating, (iii) monitoring system performance, and (iv) creating engineering designs. 5.2.2 Demonstrate proficiency in using discipline-specific tools
5.3 Demonstrate an ability to evaluate the suitability and limitations of tools used to solve an engineering problem	5.3.1 Discuss limitations and validate tools, techniques and resources 5.3.2 Verify the credibility of results from tool use with reference to the accuracy and limitations, and the assumptions inherent in their use.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.	
Competency	**Indicators**
6.1 Demonstrate an ability to describe engineering roles in a broader context, e.g. pertaining to the environment, health, safety, legal and public welfare	6.1.1 Identify and describe various engineering roles; particularly as pertains to protection of the public and public interest at the global, regional and local level
6.2 Demonstrate an understanding of professional engineering regulations, legislation and standards	6.2.1 Interpret legislation, regulations, codes, and standards relevant to your discipline and explain its contribution to the protection of the public

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and the need for sustainable development.	
Competency	**Indicators**
7.1 Demonstrate an understanding of the impact of engineering and industrial practices on social, environmental and in economic contexts	7.1.1 Identify risks/impacts in the life-cycle of an engineering product or activity 7.1.2 Understand the relationship between the technical, socio-economic and environmental dimensions of sustainability
7.2 Demonstrate an ability to apply principles of sustainable design and development	7.2.1 Describe management techniques for sustainable development 7.2.2 Apply principles of preventive engineering and sustainable development to an engineering activity or product relevant to the discipline

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice	
Competency	**Indicators**
8.1 Demonstrate an ability to recognize ethical dilemmas	8.1.1 Identify situations of unethical professional conduct and propose ethical alternatives
8.2 Demonstrate an ability to apply the Code of Ethics	8.2.1 Identify tenets of the ASME professional code of ethics 8.2.2 Examine and apply moral & ethical principles to known case studies

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.	
Competency	**Indicators**
9.1 Demonstrate an ability to form a team and define a role for each member	9.1.1 Recognize a variety of working and learning preferences; appreciate the value of diversity on a team 9.1.2 Implement the norms of practice (e.g. rules, roles, charters, agendas, etc.) of effective team work, to accomplish a goal.
9.2 Demonstrate effective individual and team operations-- communication, problem-solving, conflict resolution and leadership skills	9.2.1 Demonstrate effective communication, problem-solving, conflict resolution and leadership skills 9.2.2 Treat other team members respectfully 9.2.3 Listen to other members 9.2.4 Maintain composure in difficult situations
9.3 Demonstrate success in a team-based project	9.3.1 Present results as a team, with smooth integration of contributions from all individual efforts

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions	
Competency	**Indicators**
10.1 Demonstrate an ability to comprehend technical literature and document project work	10.1.1 Read, understand and interpret technical and non-technical information 10.1.2 Produce clear, well-constructed, and well-supported written engineering documents 10.1.3 Create flow in a document or presentation - a logical progression of ideas so that the main point is clear
10.2 Demonstrate competence in listening, speaking, and presentation	10.2.1 Listen to and comprehend information, instructions, and viewpoints of others 10.2.2 Deliver effective oral presentations to technical and non-technical audiences
10.3 Demonstrate the ability to integrate different modes of communication	10.3.1 Create engineering-standard figures, reports and drawings to complement writing and presentations 10.3.2 Use a variety of media effectively to convey a message in a document or a presentation

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments. .	
Competency	**Indicators**
11.1 Demonstrate an ability to evaluate the economic and financial performance of an engineering activity	11.1.1 Describe various economic and financial costs/benefits of an engineering activity 11.1.2 Analyze different forms of financial statements to evaluate the financial status of an engineering project
11.2 Demonstrate an ability to compare and contrast the costs/benefits of alternate proposals for an engineering activity	11.2.1 Analyze and select the most appropriate proposal based on economic and financial considerations.
11.3 Demonstrate an ability to plan/manage an engineering activity within time and budget constraints	11.3.1 Identify the tasks required to complete an engineering activity, and the resources required to complete the tasks. 11.3.2 Use project management tools to schedule an engineering project, so it is completed on time and on budget.

PO 12: Life-long learning: Recognise the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change..	
Competency	**Indicators**
12.1 Demonstrate an ability to identify gaps in knowledge and a strategy to close these gaps	12.1.1 Describe the rationale for the requirement for continuing professional development 12.1.2 Identify deficiencies or gaps in knowledge and demonstrate an ability to source information to close this gap
12.2 Demonstrate an ability to identify changing trends in engineering knowledge and practice	12.2.1 Identify historic points of technological advance in engineering that required practitioners to seek education in order to stay current 12.2.2 Recognize the need and be able to clearly explain why it is vitally important to keep current regarding new developments in your field
12.3 Demonstrate an ability to identify and access sources for new information	12.3.1 Source and comprehend technical literature and other credible sources of information 12.3.2 Analyze sourced technical and popular information for feasibility, viability, sustainability, etc.

PSO1. Professional skills: An ability to apply the knowledge in Electronics and Communication Engineering in various areas, like Communications, Signal processing, VLSI and Embedded Systems. **.**.	
Competency	**Indicators**
PSO1.1 Demonstrate an ability to apply knowledge in Communications, Signal Processing , VLSI and Embedded systems	PSO1.1.1 Design circuitry and systems related to Communications, Signal Processing, VLSI and Embedded systems PSO1.1.2 Provide solution to the problems related to the specified systems PSO1.1.3 Upgrade the systems with latest technology and incorporating more facilities

PSO2. Competency: An ability to qualify at the State, National and International level competitive examinations for employment, higher studies and research	
Competency	**Indicators**
PSO2.1 Demonstrate an ability to qualify at the State, National and International level competitive examinations for employment, higher studies and research	PSO2.1.1 Qualify top level competitive examinations for employment PSO2.1.2 Qualify top level competitive examinations for higher studies and research

Rubrics for Percentage of Performance Indicator and Mapping Grade in CO-PO matrix

Mapping ratio	Strength
>66%	**3**
<66% but >= 33%	**2**
<33% but >= 0%	**1**

CO-PO Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	3	3	---	---	---	3	2	3	2	3	1
CO2	3	1	3	2	3	---	---	---	3	2	1	3	3	3
CO3	3	---	1	---	2	---	---	---	2	1	1	---	3	3
CO4	---	3	---	3	---	---	---	---	2	---	3	---	---	---
CO5	---	---	---	---	---	---	---	---	---	---	---	---	---	---
CO6	3	2	2	1	3	---	1	---	2	2	2	3	3	1
CO7	2	2	2	3	3	---	---	---	3	2	3	2	3	1
CO8	3	1	3	2	3	---	---	---	3	2	1	3	3	3
CO9	3	---	1	---	2	---	---	---	2	1	1	---	3	3
CO10	---	3	---	3	---	---	---	---	2	---	3	---	---	---
CO11	---	---	2	1	---	2	2	---	---	---	---	---	1	---

Program level Course-PO matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO	2	2	2	2	2	2	1	---	2	1	2	2	2	2

Project Planning & Finance:

- ***July 2023***: Initially, the group discussed the possible biomedical-based projects with the mentor which could be implemented. The experiments which were already available were presented with some modifications.
- ***August 2023***: Two projects were shortlisted and their literature survey was done. By the end of the month, the project with the maximum scope of viability was finalized.
- ***September 2023***: Detailed literature survey was done and the component list was created.
- ***October 2023***: Next, we proceeded with the block diagram, flowchart and the possible algorithms which may be used.
- ***November 2023***: We worked upon the algorithm and documented the work done till date.
- ***January 2024***: We started with our hardware circuit.
- ***February 2024:*** Our algorithm was in progress. A few changes were made in the hardware circuit.
- ***March 2024:*** Were done with the algorithm and finally the hardware and software parts were integrated together.
- ***April 2024:*** Any challenges faced were overcome and finally started working on the jacket.
- ***May 2024***: Documented the whole project.

Sl.No.	Components	Quantity	Cost
1.	Sensor Pads	6	Rs. 90
2.	IC AD8232	2	Rs. 800
3.	Arduino Uno	1	Rs. 300
4.	3 - gang switch	2	Rs. 20
5.	Potentiometer	2	Rs. 300
6.	3.5mm Aux Cable	1	Rs. 50
		Total	**Rs. 1560**

Acknowledgement

The project has seen contributions from various individuals. We would like to express our special thanks to the faculties of the Department of Electronics & Communication Engineering of St. Thomas' College of Engineering & Technology, whose constant support and encouragement helped us to learn about our course more practically. We would like to extend our gratitude towards our mentor, Prashnatit Pal for helping us with the valuable suggestions to complete our project successfully.

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Ankita Singh (12200320021)

.....
Anwesha Naskar (12200320042)

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Aditya Vishwakarma (12200320056)

Dated:

Certificate

Department of Electronics and Communication Engineering

St. Thomas' College of Engineering and Technology

This is to certify that the project entitled **"CardioSens: An Advanced Technology for Heart Abnormalities Detection and Diagnosis",** has been carried out by

Ankita Singh (University Roll No.: 12200320021),
Anwesha Naskar (University Roll No.: 12200320042),
Aditya Vishwakarma (University Roll No.: 12200320056)

under my guidance during the year **July 2023 to May 2024** and accepted for partial fulfillment of the requirement of the Degree of Bachelor of Technology of Maulana Abul Kalam Azad University of Technology, West Bengal.

…….
Prashnatit Pal

Dated:

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Abstract

In the evolving healthcare landscape, technology-driven wireless sensing nodes have become the foundation of scientific knowledge. Concerns over unexpected mortality, particularly related to heart problems, have prompted the need for timely medical interventions. This report explores CardioSens, an innovative technology revolutionizing the detection and diagnosis of heart abnormalities amidst the global prevalence of cardiovascular diseases. Leveraging advanced sensors like AD8232, machine learning, and real-time analytics, CardioSens provides a non-invasive, user-friendly solution for not only continuous heart health monitoring but also the diagnosis and determination of the heart abnormalities with the help of various algorithms. The report delves into CardioSens' technical aspects, including sensor technology, data algorithms, and showcasing its efficacy compared to traditional methods. CardioSens is aimed to emerge as a promising advancement poised to positively impact healthcare by enhancing patient outcomes and contributing to the holistic well-being of those at risk of cardiovascular diseases.

Keywords: AD8232, machine learning, remote patient monitoring

Introduction

Heart diseases have emerged as a major global public health issue, contributing to a substantial number of untimely fatalities and instances of disability. Timely identification and assessment of heart conditions can significantly enhance patient results and alleviate the strain on healthcare systems. Therefore, the evolution of an effective heart diseases detection system holds immense potential in improving healthcare and saving lives.

1.1 Project Overview:

CardioSens represents a cutting-edge wearable technology with the primary objective of enhancing the early identification and diagnosis of heart conditions. The project's core aim is to develop a sophisticated jacket embedded with an array of sensors capable of constant monitoring of diverse physiological indicators. It will deliver real-time data to both users and healthcare practitioners. The seamless incorporation of these sensors into a comfortable and inconspicuous garment offers an accessible and non-intrusive method for monitoring heart health.

1.2 Objectives:

- Create a jacket with integrated sensors for the continuous monitoring of critical physiological indicators associated with heart well-being.
- Set up an intuitive interface for visualizing data and real-time monitoring.
- Incorporate algorithms and analytical tools to identify irregularities in heart function.
- Establish a secure and interoperable system for storing and transmitting data.
- Enhance the ability to promptly detect and diagnose heart diseases and various cardiac ailments.
- Enhance patient outcomes and quality of life by empowering proactive healthcare interventions.

1.3 Applications:

The CardioSens Jacket, equipped with wearable sensors for heart disease detection and diagnosis, has a wide range of potential applications across healthcare and beyond:

- Early Disease Detection
- Remote Monitoring
- Preventive Healthcare
- Elderly Care
- Research and Data Collection

Background of the Project

2.1 Literature Survey:

In 1958, a pediatric cardiologist, F. Mason Sones (1918–1985), a pediatric cardiologist at the Cleveland Clinic, pioneered a method for generating high-quality diagnostic images of the coronary arteries. This innovative procedure marked the inaugural instance where an accurate diagnosis of coronary artery disease became achievable.

Sl.No.	Paper Title	Paper Authors	Description	Drawbacks
1.	Heart Disease Detection by using Machine learning algorithms and a real time Cardiovascular Health Monitoring System. [1]	Shadman Nashif, Md. Rakib Raihan, Md. Rasedul Islam, Mohammad Hasan Imam [World Journal of Engineering and Technology, 2018]	The system includes Arduino Uno, Heartbeat sensor, Temperature sensor and humidity sensor gathers real-time data, automatically stores, and updates it on a cloud server database, and triggers notifications to the doctor in case of any uncertainty. WEKA platform along with SVM algorithm is used.	Unavailability of non-dedicated server and data poses security and privacy threats.
2.	A Novel Cardiac Auscultation Monitoring System Based on Wireless Sensing for Healthcare [2]	Haoran Ren, Hailong Jin, Chen Chen, Hemant Ghayvat,Wei Chen [IEEE Volume 6, 2018]	The system combines the acquisition, storage, and analysis of heart sounds through the utilization of the Bluetooth protocol, Hilbert-Huang transform, and sequence segmentation algorithm for physiological parameter extraction and health information interpretation.	The absence of well-optimized algorithms to yield precise outcomes and the failure to classify various cardiac conditions.
3.	Towards IoT and ML Driven Cardiac Status Prediction System. [3]	Md. Ittahad Uz Zaman, Shaira Tabassum, Md. Sabbir Ullah, Ashikur Rahaman, Samsun Nahar, A.K.M. Muzahidul Islam [ICASERT 2019]	The system employs three parameters: heart rate, ECG, and cholesterol, with the utilization of the LM358 component, Arduino Uno collects the real time data, produce the ECG Graph using AD8232 and measure the cholesterol levels using photodiode and load resistor(0-5V). The notifications are sent through an android platform using Bluetooth.	The ECG output signals were not satisfactory and the cholesterol measurement was inaccurate. Only the heart rate and cholesterol levels were visible in the mobile application.
4.	ECG Data Analysis and heart prediction using machine algorithms [4]	Sushmita Roy Tithi, Afifa Aktar, Fahimul Aleem, Amitabha Chakrabarty [Proceedings of 2019 IEEE Region 10	The researchers employed six machine learning algorithms to distinguish between normal and abnormal ECGs and to predict the probability of heart disease. The data	The accuracy score of each algorithm improved when assessing it for a specific ailment rather

		Symposium (TENSYMP)]	was split into training (75%) and testing (25%) sets, utilizing both cross-validation and a random test split approach.	than differentiating between normal or abnormal ECGs.
5.	Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques [5]	Senthilkumar Mohan, Chandrasegar Thirumalai, Gautam Srivastava [IEEE Access, Volume 7, 2019]	This study presents hybrid methods for predicting heart disease using neural networks and clinical records. It uses a radial basis function network dataset for training and 30% for classification. The Generalized Adversarial Network method diagnoses heart disease using association rules. The study also introduces a Artificial Neural Network ANN with an accuracy of up to 89.01%.	Limited incorporation of various machine learning techniques for prediction and under-developed feature selection methods for better heart disease prediction performance.
6.	Telehealth Patient Monitoring System [6]	Pamarthi Kanakaraja, K.Krishna Vamsi, I.Pratyush, M.Rama Krishna, B.Sai Mohan [IJRTE, Volume-8, Issue-4, November 2019]	The system tracks a patient's health by utilizing Heart Rate, Body Temperature, and ECG sensors. It makes use of Raspberry Pi, Arduino Mega, and the MCP3008 ADC converter to collect this data and then transmits it to the UbiDots web server. Additionally, the Raspberry Pi controller is employed for interfacing with the camera interface and establishing a Wi-Fi connection to monitor the patient's health.	The system is complex to use, requires additional assistance for handling and calculating expenses, and takes over one minute to achieve the correct result.
7.	A Study and Application Development on Monitoring Cardio-Vascular Attack using Internet of Thing (IoT) [7]	Faheem Akhtar, Anum Shakeel, Ahsan Nazir, Ahsan Wajahat, Faheem Ullah and SirajUddin Qureshi [2020 3rd International Conference on Computing, Mathematics and Engineering Technologies (iCoMET)]	The system employs a Heart Rate Sensor to observe the pulse in the earlobe by emitting infrared light and analyzing absorption variants. It sends data to a cloud-based tracker via an Arduino board and Android app, allowing instantaneous data transmission to medical professionals.	The proposed APP needs to be optimized for better performance and is not gender and disease specific.

8.	Review of IOT Based Health Monitoring System. [8]	Sipra Priyadarsini, Sujata Rana, Ravina kumara, Dr.Sasmita Pani [Gandhi Institute For Technology, Bhubaneswar, Vol-10 Issue-4 No. 1 April 2020]	The paper explores IoT-based smart healthcare systems, including Raspberry Pi for monitoring patient health parameters. Cloud computing offers flexibility, automation, and low cost. The paper also discusses the concept of S Internet of Things for sustainable healthcare delivery and RFID-based healthcare monitoring systems and security requirements.	The review stands limited due to a small dataset.
9.	IoT Based Secluded HRV System for Monitoring Hypertensive Patients [9]	M. SaiKiran Yadav, M. Mahaboob Basha, B. Akhil Kumar [6th International Conference on Recent Trends on Electronics, Information, Communication & Technology (RTEICT), August 27th & 28th 2021]	The suggested system observes heart rate variability in individuals with Huntington's disease within an IoT framework. It utilizes two sensors, namely MAX30102 and DHT11, to gauge light absorption levels and humidity. The data collected from these sensors is then transmitted to an Arduino Uno, which displays it on an LCD screen. In case the data does not meet the predefined threshold, alarms are triggered, allowing for remote monitoring through ThingSpeak.	The huge size of the device is a major issue and the accuracy and precision can be improved upon.
10.	An Outcome based Analysis Heart Disease Prediction using machine learning Algorithms and Data Mining approaches. [10]	Aushtmi Deb, Mst. Sadia Akter Koli, Sheikh Beauty Akter, Adil Ahmed Chowdhury [IEEE AIIoT 2022]	The experiment involved the utilization of six chosen classifiers, including logistic regression, decision tree, random forest, naive Bayes, SVM, and KNN. Subsequently, experimentation was performed on the gathered datasets, and the outcomes were subsequently compared.	The research was limited due to a smaller and unreal dataset.
11.	Heart Disease Prediction Using Machine Learning Algorithms [11]	Akash Yadav, Dr. Vikas Jindal [2022 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON)]	The system begins with data collection and identification of relevant properties, followed by data transformation and splitting into training and testing sets. Machine learning algorithms like AdaBoost, SVM, NB, DT, XGBoost, RT, and LR are applied for disease prediction. Comparative analysis helps select a high-precision algorithm.	Utilization of small-scale databases. As the value of the database increases, there is a rise in asymptotic errors and a decrease in bias. In such cases, methods relying on low variance tend to deliver superior performance.

12.	A Novel Web-Based Multi-Class Heart Disease Prediction Using Machine Learning Algorithms [12]	Toqeer Ahmed, Saeed Mian Qaiser [2022 International Conference on Electrical Engineering and Sustainable Technologies (ICEEST)]	The system enhances the classification of heart disease by employing diverse techniques such as Random Forest, SVC, KNN, XGBoost, and ensemble learning classifiers. Performance evaluation was carried out within the FLASK framework. The Stacking ensemble classifier attained a 93.5% accuracy rate and a ROC score of 0.99%, facilitating doctors.	The project could have been created using React JS, Node JS, and a MongoDB database hosted by the FLASK APP.
13.	An Analytical Review of Heart Failure Detection based on IoT and Machine Learning [13]	Aroh Gahane, Dr. Chinnaiah Kotadi [Proceedings of the Second International Conference on Artificial Intelligence and Smart Energy (ICAIS-2022)]	Data obtained from different body positions underwent processing with a combination of deep learning, machine learning, and conventional classification algorithms, which included SVM, Adaboost, Logistic Regression, and kNN.	Accuracy can be improved upon using various other optimized algorithms.
14.	Heart Diseases Prediction Using Machine Learning Algorithms [14]	E. Anupriya, Manyam Thaile, P.Chinnasamy, M. Laxmi Narayana [2023 International Conference on Computer Communication and Informatics (ICCCI), Jan. 23 – 25, 2023, Coimbatore, INDIA]	In the research, machine learning algorithms such as Decision Tree, KNN, Naive Bayes, and Random Forest are employed for the prediction of cardiac disease. The study involves an evaluation of their sensitivity and the comparison of various types, including Logistic Regression, Random Forest, Naive Bayes, and SVM.	Optimized algorithms with more accurate results were not found.
15.	IoT Based Heart attack Detection and Heart Rate Monitoring System [15]	Dr. T. Arun Prasath, M.Mohamed Arif, S Srinivasan, A Muthumanojkumar, M Sushmitha, Sakthivel Sankaran [2023 Advanced Computing and Communication Technologies for High Performance Applications (ACCTHPA) , IEEE] |	The prototype uses pulse and ECG sensors to measure heart rate and electrical activity, processing signals via Arduino Uno and tracking the patient via GPS module.	Size of the wearable may create an issue to the user and hence in portability of the device.

2.2 Research Gap:

The literature survey suggests opportunities for future studies to focus on improving the security, privacy, accuracy, usability, and performance of IoT-based healthcare systems, as well as exploring more effective algorithms and addressing challenges related to data, devices, and system integration.

Problem Statement

Lack of user-friendly, comfortable, and continuously monitoring wearable technology that provides real-time health information with increased security, privacy, and accuracy.

Details of the Project

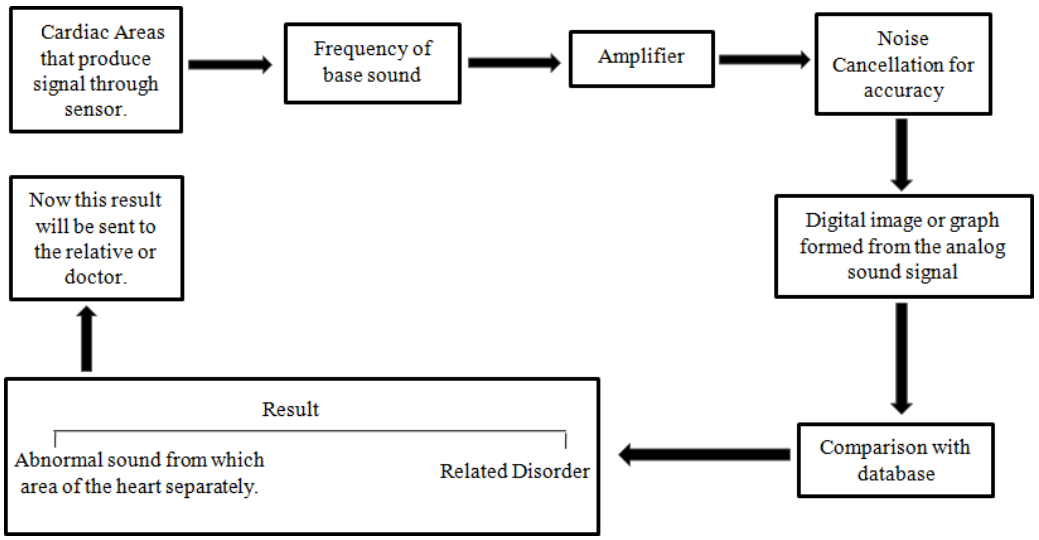
3.1. Description:

CardioSens is a state-of-the-art wearable technology designed to help detect heart problems early. The main goal of this project is to create a comfy jacket various sensors like heart rate monitor, temperature sensor and pressure sensor, that can continuously keep a check on an individual's health. It will show the information in real-time to both the user and their doctor or relatives registered.

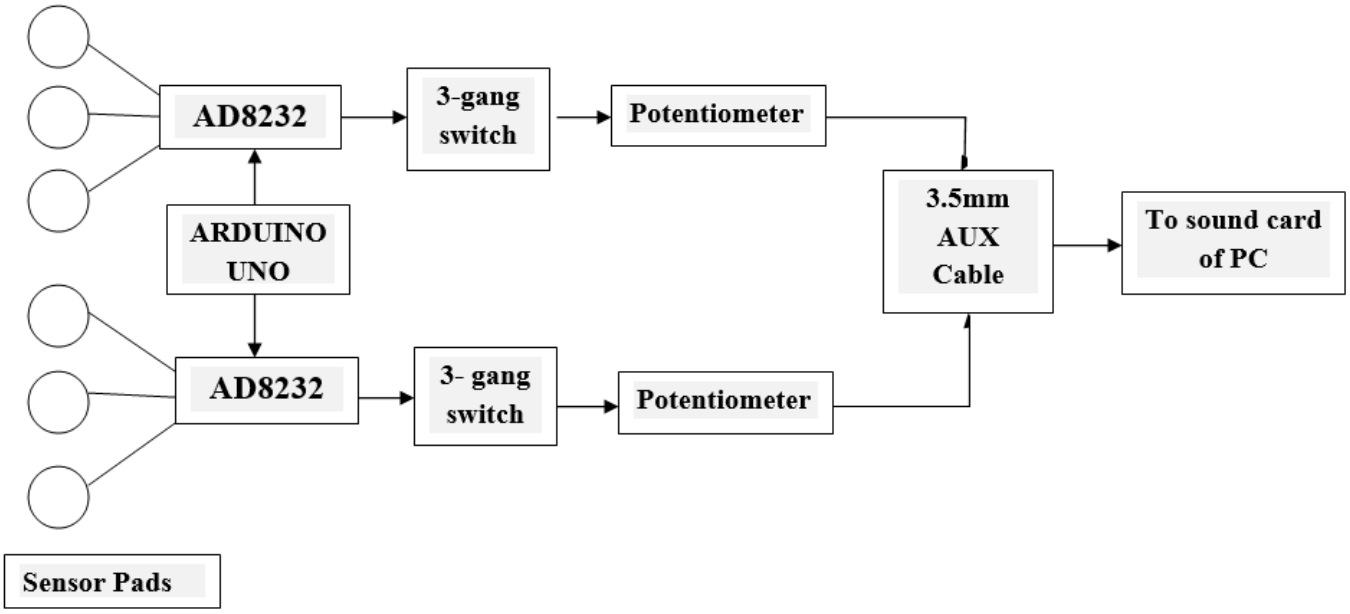
3.2. Component List:

Sl.No.	Components	Quantity
1.	Sensor Pads	6
2.	IC AD8232	2
3.	Arduino Uno	1
4.	3 - gang switch	2
5.	Potentiometer	2
6.	3.5mm Aux Cable	1

3.3. Flowchart:



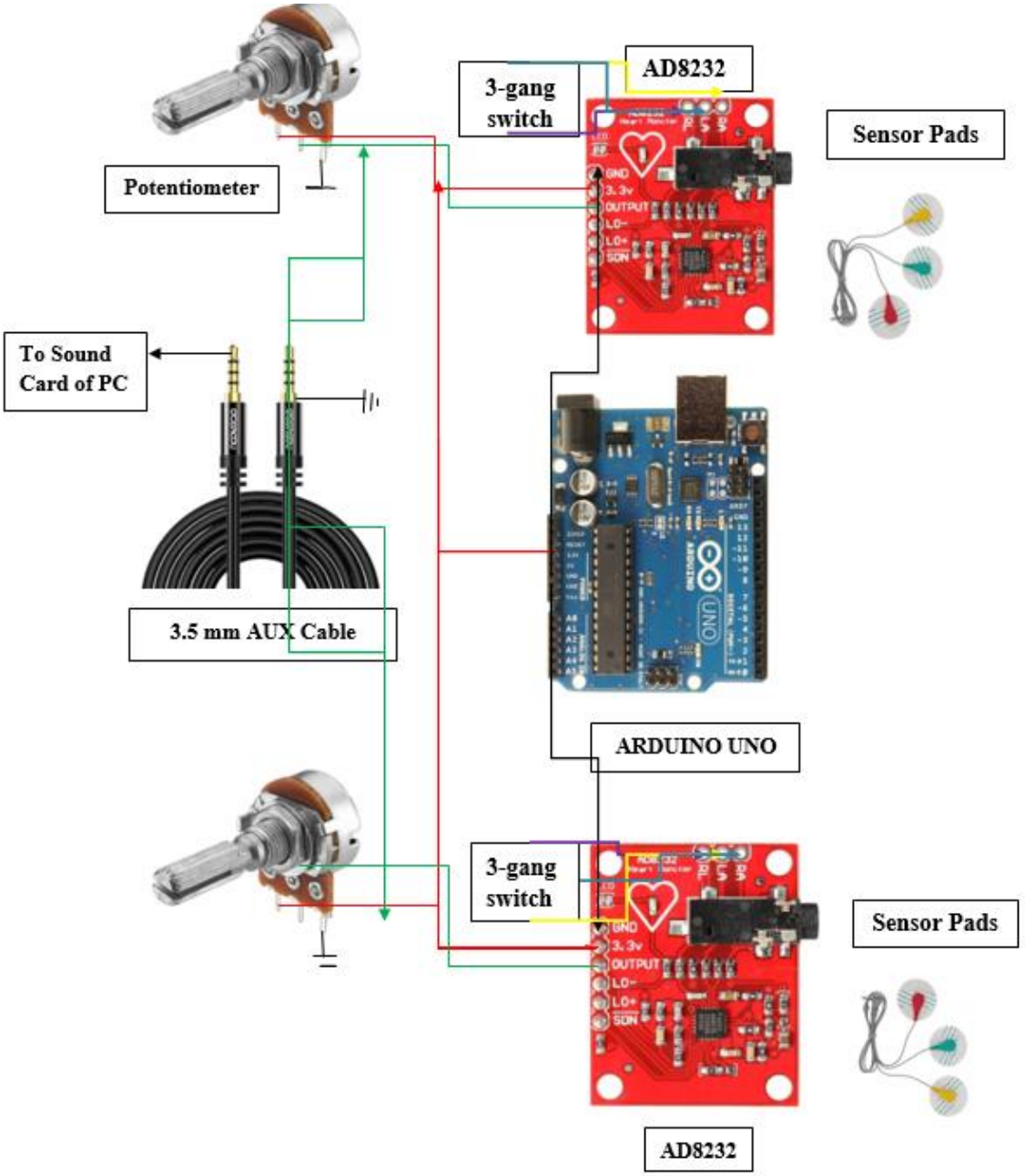
3.4.Block Diagram:



Explanation of Block Diagram:

- **Sensor Pads:** These are electronic devices crafted for the purpose of identifying or quantifying physical factors like pressure, temperature, humidity, movement, or proximity. In our project, we employ them to identify cardiac signals.
- **AD8232**: The AD8232 serves as an integrated signal conditioning component for ECG and similar biopotential measurement tasks. Its purpose is to extract, enhance, and filter minor biopotential signals even when surrounded by disruptive factors like motion or the placement of electrodes at a distance.
- **Arduino Uno:** The Arduino Uno is a microcontroller board that utilizes the ATmega328P as its core component. It is equipped with digital input/output pins, analog inputs, a USB interface, and additional functionalities that enable it to connect with a wide range of sensors, actuators, and electronic elements. An Arduino Uno is employed to transmit cardiac monitor data to the Raspberry Pi.
- **3-gang switch:** A switch that incorporates three individual switches in a single unit. Also known as triple-gang switch. This type of switch assembly contains three separate switches within one wall plate, allowing control of three different electrical circuits or devices (in our case, signals) from a single location.
- **Potentiometer:** A potentiometer, also known as a pot, is a three-terminal resistor with a rotating contact that forms an adjustable voltage divider. It functions by varying the resistance in a circuit manually, typically by turning a knob or sliding a slider.
- **3.5 mm Aux Cable**: An aux cable is being used to connect the hardware components to the processor by connecting it to the sound card of the laptop.

3.5.Circuit Diagram:



3.6. Specifications:

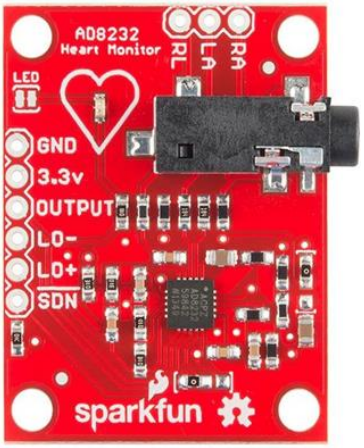


Fig.1. IC AD8232

PIN CONNECTIONS

The AD8232 ECG Module Monitor breaks out of nine IC links. The module consisting of three ECG Pads Those is Black Acts as a Common Node, Blue acts as a LA (Left Arm) and Red Acts as RL (Right Leg).

IC AD8232 ECG	Pinouts	Connection with Arduino	Connection with Potentiometer
GND	Ground	GND	GND
3.3V	3.3V Power Supply	3.3V	Vcc
OUTPUT	Output Pin	Not used	OUTPUT
LO-	Leads Off Detect Minus	Not used	Not used
LO+	Leads Off Detect Plus	Not used	Not used
SDN	Shut down	Not used	Not used

Table-1: Pin Connections

For optimal measurements, the chest pads should be placed close together. Using conductive jelly can improve the body's response time. Color-coded cables help ensure the correct orientation based on Einthoven's triangle.

Cable Color	Signal
Black	RA (Right Arm)
Blue	LA (Left Arm)
Red	RL (Right Leg)

Table-2: Sensor placement details

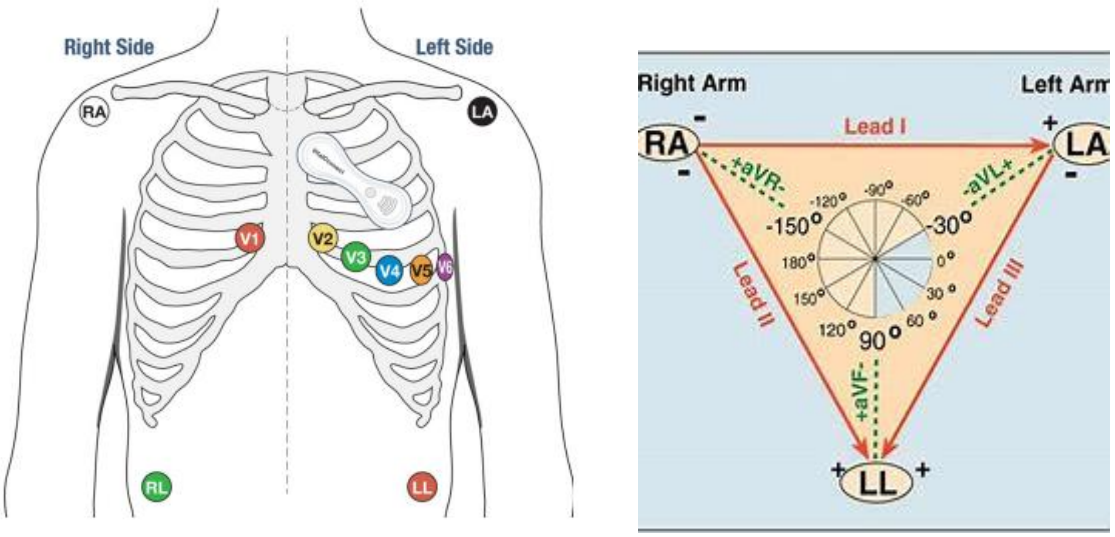


Fig.2. Layout of electrode placement of Vital Patch sensor and 12-lead ECG device in Mason-Likar position.

Formula to get ECG signals:

Leads	ECG Signals
I	LA - RA
II	LL - RA
III	LL - LA
aVR	RA – (RA+ LA+LL)/3
aVL	LA – (RA+ LA+LL)/3
aVF	LL – (RA+ LA+LL)/3
V1	C1 – (RA+ LA+LL)/3
V2	C2 – (RA+ LA+LL)/3
V3	C3 – (RA+ LA+LL)/3
V4	C4 – (RA+ LA+LL)/3
V5	C5 – (RA+ LA+LL)/3
V6	C6 – (RA+ LA+LL)/3

Graphs obtained through single leads:

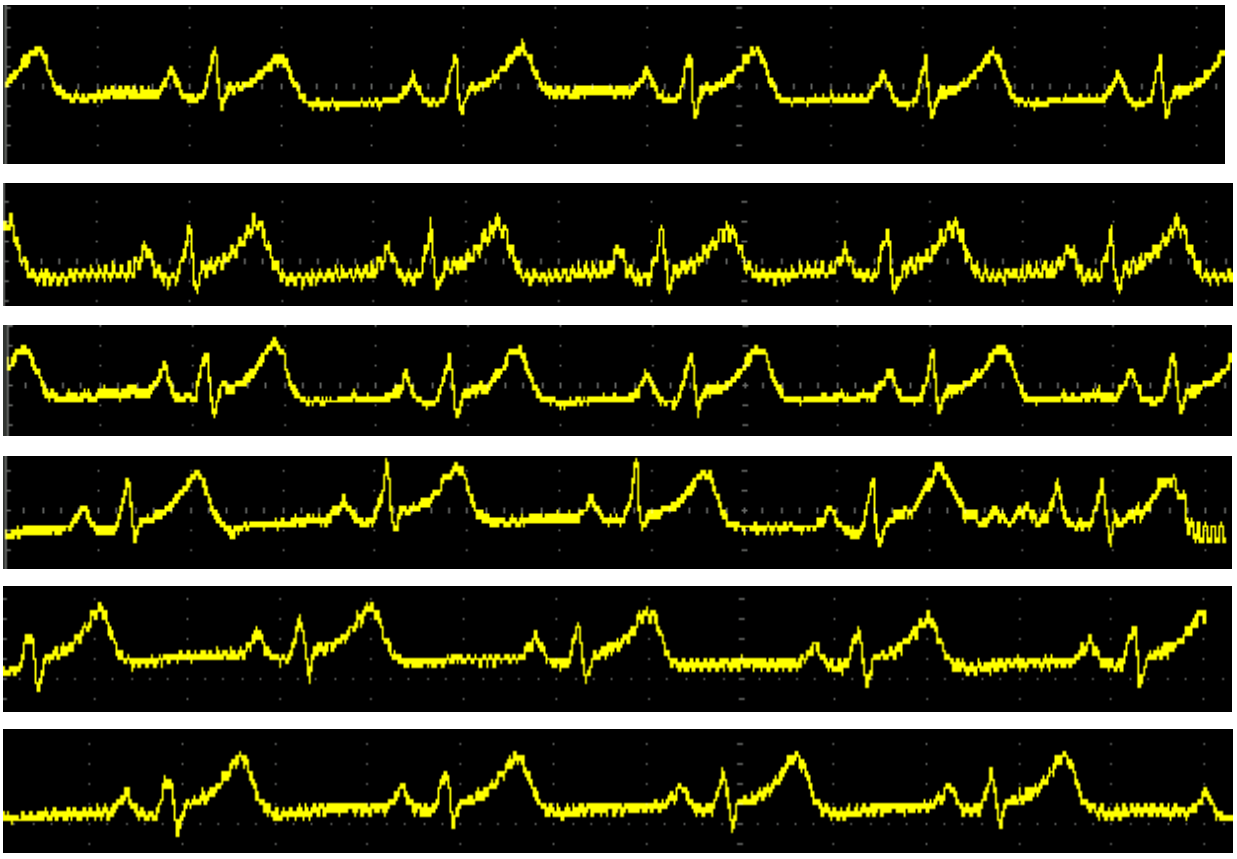


Fig.3.Representative ECG signal from leads I, II, III, aVR, aVL, aVF

3.7 Algorithm:

Image Preprocessing:

1.1. Load the image from image_path using cv2.imread in grayscale mode.

1.2. If the image is successfully loaded:

- Resize the image to (256, 256) with cv2.resize.
- Normalize the image to the range [0, 1] by dividing it by 255.0.
- Convert the normalized image to 8-bit representation by multiplying by 255 and casting to uint8.
- Apply histogram equalization with cv2.equalizeHist.
- Save the enhanced image to a predefined output path.

1.3. If image loading fails, print an error message.

ECG Signal Processing:

2.1. Load ECG data from data_path with np.loadtxt.

2.2. If data loading is successful:

- Process the ECG signal with ecg.ecg, retrieving R-peaks and other information.
- For each R-peak:
- Define a window around the R-peak to extract the PQRS-T complex.
- Calculate features based on the complex:
- Amplitudes (P, QRS, T).
- Durations.
- Slopes.
- Save the extracted complex and features to a separate .npy file.
- Save all extracted features to "all_features.npy".

2.3. If data loading fails, print an error message.

Main Workflow:

3.1. Define image_path and data_path.

3.2. Execute the process_image function with image_path.

3.3. Execute the process_ecg_signal function with data_path.

3.4. Print "Processing complete." when done.

Considerations and Potential Improvements:

- **Error Handling:** Improve error handling for I/O operations and edge cases.
- **Optimization:** Consider optimizing the loop for feature extraction, especially with larger ECG datasets.

- **Validation:** Add validation checks to ensure the ECG data is in the expected format before processing.
- **Logging:** Introduce logging mechanisms for better traceability during processing.
- **Extendibility:** Structure the code in a more modular manner for future enhancements or changes in processing logic.

The final ECG signal on which the algorithm works upon to diagnose the disorder is as follows:

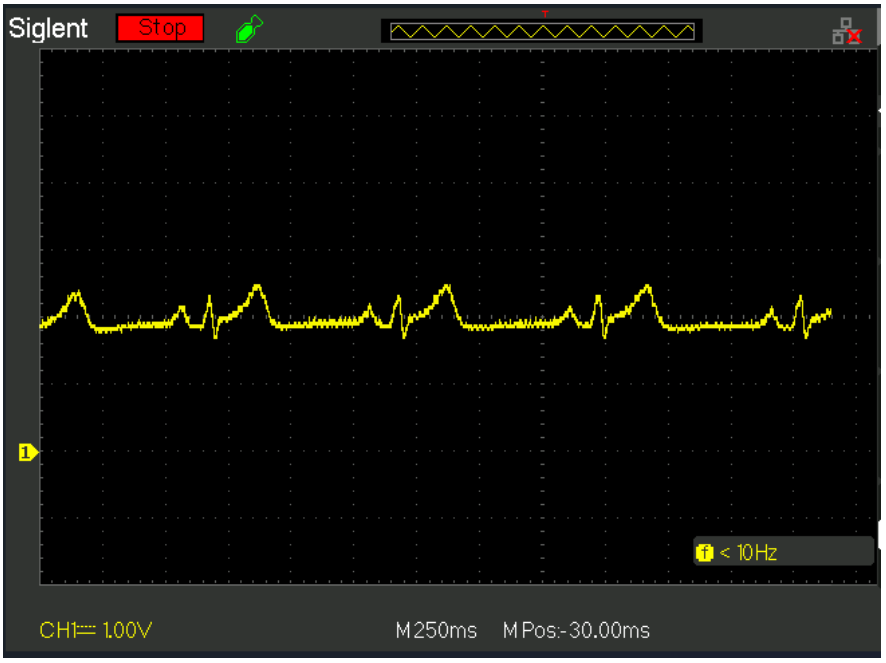


Fig.4. Processed ECG signal

After the whole processing is complete, an email notification is sent to the authorized doctor or relative.

```
import smtplib

email = input("sender email: ")
receiver_email = input("receiver email: ")

subject = input("subject: ")
message = input("message: ")

text = f"Subject: {subject}\n\n{message}"

server = smtplib.SMTP("smtp.gmail.com", 587)
server.starttls()

server.login(email, "qezh wyzv slcp pulc")

server.sendmail(email, receiver_email, text)

print("email has been sent to " + receiver_email)
```

Fig.5. Program for email notification.

Results

4.1 Result Analysis:

The following figure depicts the hardware connection.

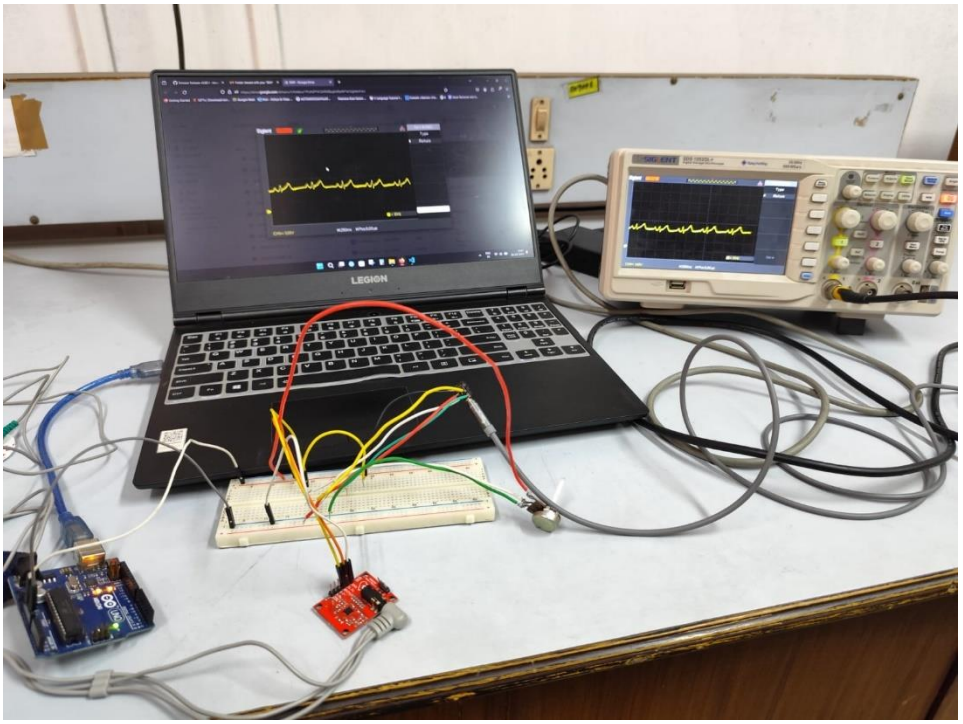


Fig.5. Hardware Connection

The graph generated is worked upon by the above-mentioned algorithm and email notifications are sent.

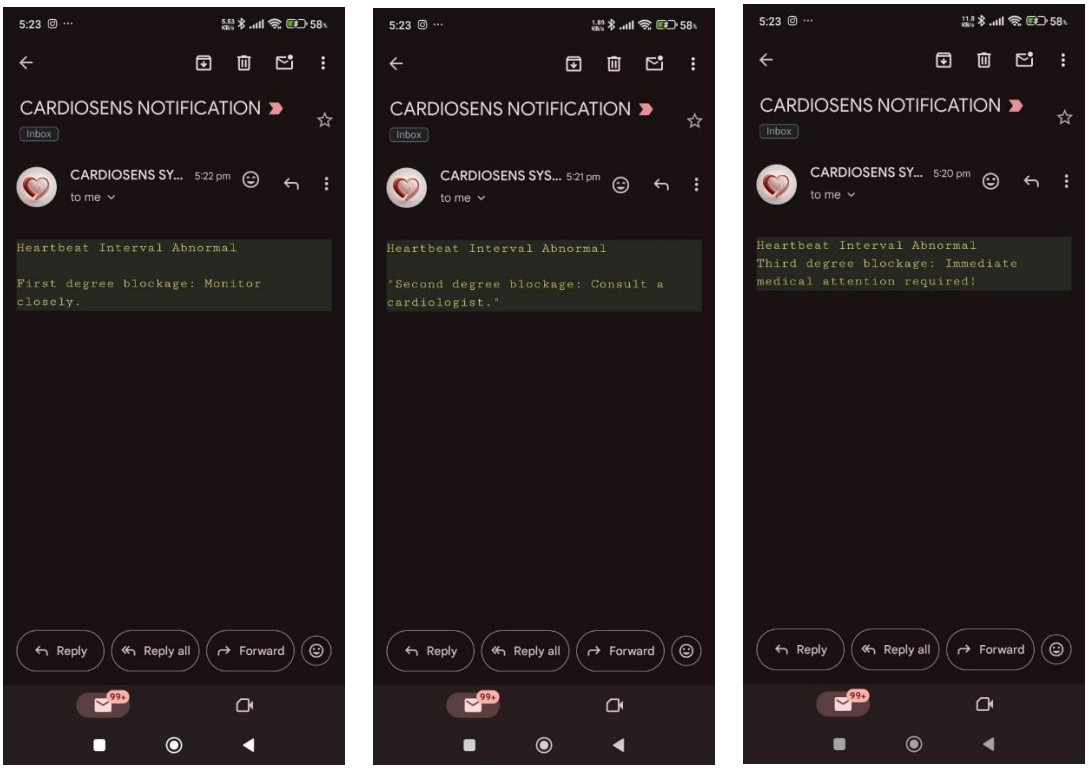


Fig.6. Email notifications (3 cases)

We have worked upon 15 heart abnormalities mentioned below:

- **Atrial Fibrillation (AFib)**: ECG shows irregularly irregular rhythm with absent P waves and rapid, irregular QRS complexes.

- **Myocardial Infarction (Heart Attack):** ST-segment elevation or depression in specific leads corresponding to the area of myocardial damage. T-wave inversion in the affected leads.

- **Ventricular Fibrillation:** Chaotic, irregular electrical activity with no discernible P, QRS, or T waves.

- **First degree heart block:** PR interval > 0.22s

- **Second degree heart block:** R peak < 0.8mV

- **Third degree heart block:** P peak if almost 0 or infinity

- **Wolff-Parkinson-White Syndrome:** Shortened PR interval (<0.12 seconds) with a delta wave, indicating early ventricular depolarization.

- **Bundle Branch Blocks:**
- **Right Bundle Branch Block (RBBB):** Widened QRS complex (>0.12 seconds) with a characteristic RsR' pattern in V1 and V2.
- **Left Bundle Branch Block (LBBB):** Widened QRS complex (>0.12 seconds) with broad, notched or slurred R waves in lateral leads (I, aVL, V5, V6) and deep S waves in right precordial leads (V1, V2).

- **Long QT Syndrome**: Prolonged QT interval (>0.44 seconds) on the ECG.

- **Pericarditis:** Diffuse ST-segment elevation across multiple leads, often with PR depression.

- **Hypertrophy:**
- **Left Ventricular Hypertrophy (LVH):** Increased voltage in the limb leads (S wave in lead V1 + R wave in lead V5 or V6 ≥ 35 mm), and/or presence of left-axis deviation, and/or repolarization abnormalities.
- **Right Ventricular Hypertrophy (RVH):** Increased amplitude of the R wave in lead V1 (R/S ratio >1), along with right axis deviation and right atrial enlargement.

- **Bradycardia:** Heart rate <60 beats per minute, with normal P waves and QRS complexes.

- **Tachycardia:** Heart rate >100 beats per minute, with normal or abnormal P waves and QRS complexes depending on the type of tachycardia.

Comments on the results

The heart disease detection and diagnosis project, which presently utilizes an advanced algorithm to identify eight different diseases, demonstrates significant potential in improving early diagnosis and personalized treatment plans, potentially leading to better patient outcomes and more efficient healthcare delivery.

Conclusion

In conclusion, the CardioSens Jacket is a groundbreaking innovation that has the potential to transform the way we address the detection and diagnosis of heart diseases. By seamlessly integrating advanced sensor technology into an everyday, comfortable garment, this project opens new possibilities for early detection and continuous monitoring of heart conditions. The applications of this wearable technology extend to various areas in healthcare, well-being, and beyond, offering the prospect of better patient outcomes, proactive interventions, and a more informed approach to heart health.

Future Works

The sensor pads may be replaced by wireless sensors that may offer greater flexibility, ease of installation, reduced clutter, enhanced safety, and better real-time data transmission compared to sensor pads. The algorithm may be updated to include a greater number of diseases and hence lead to better accuracy.

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